

Failure ledger of the block-1 study

Which settings and datasets break, through which mechanism,
and what each failure does to the Brier score and the CRPS

Technical note

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Abstract

This note is the consolidated ledger of every anomaly observed in the block-1 positive control (settings 4–5, methods i.i.d. and AR(2), datasets 1–10, train $t = 1..50$, forecast $t = 51..65$). Each affected cell is classified into one of four pathologies — sign reflection (R), positive-branch inflation (P), data-level under-identification (U), and heavy-tail realization (H) — with its mechanism stated as a formula and its measured effect on the two forecast scores. The summary judgment, which the companion notes quantify in full, is that only unrepaired reflections with $|\hat{\lambda}_0| \gtrsim 5$ move the scores materially (Brier up to $12\times$); every other pathology leaves the paired score comparison intact.

1 Model, ridge, and scores

The fitted model, $K = 1$ knot, is

$$y_{st} = \beta_0 + x'_s \beta + \lambda z_t + \varepsilon_{st} / \sqrt{\tau_t}, \quad z_t \geq 0, \quad z_t \sim \text{HN}(0, \tau_t^{-1/2}) \text{ with AR(2) copula,} \quad (1)$$

with simulation truth $\lambda = 3$, $\beta_0 = 10$, internal τ -hyperparameters $(a, b) = (3, 8)$, and shared latent AR(2) coefficients $\phi = (0.80, -0.35)$ in setting 4 and $(0.15, 0.80)$ in setting 5 (spectral radii 0.592 and 0.973; memory horizons $h^* = 6$ and 109 days).

The likelihood sees (β_0, λ, z) only through $m_t = \beta_0 + \lambda z_t$. For any (β_0^*, λ^*) the reassignment

$$z_t^* = \frac{m_t - \beta_0^*}{\lambda^*}, \quad z_t^* \geq 0 \quad \forall t, \quad (2)$$

reproduces the likelihood exactly. The flat family (2) has a positive branch ($\lambda^* > 0$, path shape kept) and a reflected branch ($\lambda^* < 0$, path flipped, $\beta_0^* > \max_t m_t$). Identification along the ridge comes only from the prior on z .

Scores are computed per cell on the validation block, means over leads 1–5 (the pre-registered primary window). CRPS is the sample CRPS over the posterior predictive draws; the Brier score uses the q_{95} threshold of the cell's own validation block.

2 The ledger

Table 1 lists all 13 cells whose attempt-0 chain failed the (pre-registered) assertions under the symmetric prior $\lambda \sim N(0, 20)$, together with their fate under the two remedies that were subsequently tested, and the three datasets whose fitted τ -shape falls below the $a = 2$ moment boundary of Section 3. Pathology codes:

- R reflection:** chain captured on the $\lambda < 0$ branch of (2) during burn-in;
- P positive inflation:** same ridge, positive branch ($\hat{\lambda}$ too large, z compressed, sign correct);
- U under-identification:** the 50-day training window realized a nearly flat z path, so λ is not identified by the data (stable across four seeds and both priors);
- H heavy-tail realization:** fitted $a < 2$, so the predictive-spread statistic has no finite sampling variance (a property of the dataset, not of any chain).

Table 1: All affected cells. $\hat{\lambda}_0$ = attempt-0 posterior mean under $N(0, 20)$; “guard” = assertion-guarded reseeding (att = attempts used, 4 = flagged); “HN” = single fit under $\lambda \sim \text{HN}(0, 20)$, same seed, no guard. B/C = truth-recovery and spread assertions, recorded descriptively. Scores are Brier q_{95} , leads 1–5.

cell		type	$\hat{\lambda}_0$	guard (att)	HN: $\hat{\lambda}$, B/C	Brier q_{95}		
						att-0	guard	HN
s4 d5 m2	AR(2)	R	−7.28	+3.93 (2)	+3.56, ✓✓	.0205	.0017	.0016
s4 d5 m1	i.i.d.	R	−7.23	+3.52 (2)	+3.56, ✓✓	.0265	.0033	.0036
s4 d10 m1	i.i.d.	R	−4.97	+2.99 (2)	+2.84, ✓✓	.0558	.0226	.0184
s5 d7 m1	i.i.d.	R	−3.99	+2.47 (2)	+2.52, ✓✓	.0832	.0463	.0465
s4 d4 m2	AR(2)	R	−4.01	+2.19 (2)	+2.12, ✓✓	.0010	.0022	.0019
s4 d6 m2	AR(2)	R+H	−8.90	+2.86 (3)	+3.23, ✓×	.1378	.1349	.1338
s5 d5 m1	i.i.d.	R	−4.28	+2.60 (2)	+2.58, ✓✓	.0164	.0166	.0166
s5 d6 m2	AR(2)	P	+5.40	+3.95 (2)	+2.92, ✓✓	.0794	.0799	.0808
s5 d8 m1	i.i.d.	R	−2.62	+2.29 (4)	+2.27, ✓✓	.0000	.0007	.0005
s5 d3 m2	AR(2)	R	−15.18	+3.34 (4)	+4.97, ×✓	.1088	.1323	.1282
s5 d2 m1	i.i.d.	U	+0.23	flag (4)	+0.29, ×✓	.0194	.0194	.0194
s5 d2 m2	AR(2)	U	+0.55	flag (4)	+0.57, ×✓	.0190	.0190	.0190
s5 d8 m2	AR(2)	R	−2.47	flag (4)	+3.63, ✓✓	.0000	.0000	.0005
<i>heavy-tail datasets</i> (property of the data; both methods, both priors agree on $\hat{a}$ to two decimals):								
s4 d1	both	H	$\hat{a} \approx 1.95$	(truth $a = 3$)				
s4 d6	both	H	$\hat{a} \approx 1.51$	(→ the one C exceedance, see §3)				
s5 d5	both	H	$\hat{a} \approx 1.93$					

Reflections struck both methods at comparable rates (6 of 13 affected cells are the i.i.d. method), which established early that the pathology belongs to the shared skew-t family of (1), not to the AR(2) time structure.

3 Mechanisms

R — reflection. Once a chain drifts onto the $\lambda^* < 0$ branch of (2) during burn-in, escape requires simultaneously moving λ across zero, dropping β_0 by ≈ 20 , and re-inverting the entire z path; every one-coordinate Gibbs or MH move is individually catastrophic for the likelihood, so the posterior is bimodal and the chain cannot switch. Capture is decided by burn-in randomness, which is why reseeding cured 10 of 13 cells.

The $1/|\lambda|$ compression and the forecast law. On the ridge $\text{sd}_t(z) = \text{sd}_t(m)/|\lambda|$, so an inflated loading forces an over-smooth path. Forecasting redraws z from the model law at full

marginal variance, which yields the empirical law

$$\max_h \text{sd}(\hat{y}_h) \approx 1.49 |\hat{\lambda}| \quad (R^2 = 0.987 \text{ through the origin}), \quad (3)$$

the single relation that turns a hidden fitting inconsistency into a visible forecast explosion, and that makes the score damage grow with $|\hat{\lambda}_0|$.

The accomplice. The over-smooth path should be penalized by the z prior, whose path density per step is $-\varepsilon_t^2/2\sigma_\varepsilon^2(\phi_z) - \log \sigma_\varepsilon(\phi_z)$ with innovation scale $\sigma_\varepsilon(\phi_z) \rightarrow 0$ as the AR(2) roots approach the unit circle. Because ϕ_z is estimated, its update reads the compressed path as evidence of high persistence, drifts toward the unit circle, and reprises the violation as legal long memory. The dynamics parameter is the accomplice, not the trigger.

U — under-identification. Setting 5 has $h^* \approx 109$ days against a 50-day training window. A window that realizes a nearly flat z path carries no information about λ ; dataset 2 reproduced $\hat{\lambda} \approx 0.25$ (i.i.d.) and ≈ 0.56 (AR(2)) across four seeds and both priors, stable to three decimals. No sampler-side or prior-side intervention can create information the window does not contain.

H — heavy tails and the $a = 2$ boundary. With $\sigma^2 = 1/\tau \sim \text{IG}(a, b)$,

$$\mathbb{E}[\sigma^4] = \frac{b^2}{(a-1)(a-2)}, \quad (4)$$

which is infinite for $a \leq 2$. The truth $a = 3$ has finite fourth moment; three datasets nevertheless realized training windows whose fitted $\hat{a} < 2$ (Table 1). Below the boundary the sample SD of the predictive cloud is an estimator with infinite sampling variance, so it never stabilizes at any Monte-Carlo size, and a handful of iterations — each amplified by the fact that one τ_t draw scales all 144 stations at once — decide whether a spread threshold is crossed. The one C exceedance at s4 d6 m2 under HN (worst lead SD 36.9 against a threshold of 13.1) is this coin flip, not a sick chain; the four chains fitted to d6 agree in their posteriors and in every predictive quantile up to $q_{99.9}$.

4 Which pathologies touch the scores

Table 2: Score impact by pathology, leads 1–5. “If shipped” = the damage of delivering the attempt-0 chain, measured by bit-for-bit refits of every replaced chain.

type	cells	impact if shipped	impact on paired AR(2)–i.i.d.
R, $ \hat{\lambda}_0 \geq 5$	4	Brier 8–12 $\times$, CRPS +40%	material if unprotected
R, $ \hat{\lambda}_0 < 5$	6	small to none (law (3))	none measured
P	1	none (.0794 vs .0799)	none
U	2	none (scores identical)	none (symmetric across methods)
H	3 datasets	none (predictive clouds agree to $q_{99.9}$)	none

Three observations close the ledger.

1. The only score-relevant pathology is an unrepaired large reflection. Across the 13 affected cells, repairing them improved the Brier score by 16% and the CRPS by 12% on average, with the whole effect concentrated in the $|\hat{\lambda}_0| \geq 5$ rows of Table 1.
2. On difficult validation windows an inconsistent fit can *outscore* its healthy replacement (s5 d3 m2, twice, under both remedies), because overdispersion hedges. Selection must therefore never be done on scores.
3. U and H are properties of the data, are symmetric across the compared methods, and cancel in the paired comparison; they require reporting, not repair.

Companion notes — `tex/lambda_phiz_ridge` (geometry and law (3)), `tex/guard_effect` (score counterfactual of the guard), and `tex/ar2_remedy_ledger`, `tex/ar2_score_findings` (remedies; main score findings).